

Rearranging Equations - Solutions

LLE – Mathematics and Statistics Skills

ECO-5007A

Skills - Rearranging Equations

In each of the following expressions, make x the subject of the equation.

1. **Question:** $x + y = 12$

Solution: Subtract y from both sides.

$$x = 12 - y$$

2. **Question:** $5x - 4 = y$

Solution: Add 4 to both sides, then divide by 5.

$$5x = y + 4$$

$$x = \frac{y + 4}{5}$$

3. **Question:** $ax - 3by = 9$

Solution: Add $3by$ to both sides, then divide by a .

$$ax = 9 + 3by$$

$$x = \frac{9 + 3by}{a}$$

4. **Question:** $y = mx + c$

Solution: Subtract c from both sides, then divide by m .

$$y - c = mx$$

$$x = \frac{y - c}{m}$$

5. **Question:** $2 + 5xy = 6$

Solution: Subtract 2 from both sides, then divide by $5y$.

$$5xy = 4$$

$$x = \frac{4}{5y}$$

6. **Question:** $9x + 6y = 4x + 1$

Solution: Gather all terms containing x on one side and other terms on the opposite side. Then, factor out x .

$$9x - 4x = 1 - 6y$$

$$5x = 1 - 6y$$

$$x = \frac{1 - 6y}{5}$$

Tip: Gather all terms containing x on one side of the equation and all other terms on the opposite side. Then, factor out x .

7. **Question:** $ax + by = 4x + 1$

Solution: Gather x terms, factor out x .

$$ax - 4x = 1 - by$$

$$x(a - 4) = 1 - by$$

$$x = \frac{1 - by}{a - 4}$$

8. **Question:** $ax - by = bx - 8$

Solution: Gather x terms, factor out x .

$$ax - bx = by - 8$$

$$x(a - b) = by - 8$$

$$x = \frac{by - 8}{a - b}$$

9. **Question:** $\frac{2ty + mx}{ty - mx} = 1$

Solution: Multiply both sides by the denominator, then gather x terms.

$$2ty + mx = ty - mx$$

$$mx + mx = ty - 2ty$$

$$2mx = -ty$$

$$x = -\frac{ty}{2m}$$

10. **Question:** $\frac{ty + mx}{ty - 3mx} = k$

Solution: Multiply by the denominator, expand, gather x terms, factor out x .

$$ty + mx = k(ty - 3mx)$$

$$ty + mx = kty - 3kmx$$

$$mx + 3kmx = kty - ty$$

$$x(m + 3km) = ty(k - 1)$$

$$x = \frac{ty(k - 1)}{m(1 + 3k)}$$

11. **Question:** $9x^2 = 4y$

Solution: Divide by 9, then take the square root.

$$x^2 = \frac{4y}{9}$$

$$x = \pm \sqrt{\frac{4y}{9}} = \pm \frac{2\sqrt{y}}{3}$$

12. **Question:** $x^3 - 5m = t$

Solution: Add $5m$ to both sides, then take the cube root.

$$x^3 = t + 5m$$

$$x = \sqrt[3]{t + 5m}$$

13. **Question:** $x^2 - 2x = 15$

Solution: Rearrange to standard quadratic form $ax^2 + bx + c = 0$.

$$x^2 - 2x - 15 = 0$$

Factor the quadratic equation.

$$(x - 5)(x + 3) = 0$$

$$x = 5 \quad \text{or} \quad x = -3$$

Tip: Rearrange this into the standard quadratic form $ax^2 + bx + c = 0$ and then consider using the quadratic formula or factoring.

14. **Question:** $x^2 = 2 - qx$

Solution: Rearrange to standard quadratic form.

$$x^2 + qx - 2 = 0$$

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$$x = \frac{-q \pm \sqrt{q^2 - 4(1)(-2)}}{2(1)}$$

$$x = \frac{-q \pm \sqrt{q^2 + 8}}{2}$$

15. **Question:** $x^2 - 8 = kx$

Solution: Rearrange to standard quadratic form.

$$x^2 - kx - 8 = 0$$

Use the quadratic formula.

$$x = \frac{-(-k) \pm \sqrt{(-k)^2 - 4(1)(-8)}}{2(1)}$$

$$x = \frac{k \pm \sqrt{k^2 + 32}}{2}$$

- **Bonus Mark Solution:** For this expression to have real solutions, the discriminant must be non-negative: $b^2 - 4ac \geq 0$. In this case, $k^2 + 32 \geq 0$. Since k^2 is always non-negative, $k^2 + 32$ will always be greater than or equal to 32, which is always greater than 0. Therefore, this expression **always** has real solutions for any real value of k .

Rearranging with Indices

This section uses the index law of $x^a x^b = x^{a+b}$ and that a number multiplied by its reciprocal is 1. This is useful for more difficult indices.

Practice Problems:

Rearrange the following to make x the subject:

1. **Question:** $x^{\frac{3}{4}} = y^{\frac{1}{2}}$

Solution: Raise both sides to the power of $\frac{4}{3}$.

$$\left(x^{\frac{3}{4}}\right)^{\frac{4}{3}} = \left(y^{\frac{1}{2}}\right)^{\frac{4}{3}}$$

$$x = y^{\frac{4}{6}} = y^{\frac{2}{3}}$$

2. **Question:** $x^{\frac{3}{4}} = y^{\frac{1}{2}} - 2$

Solution: Raise both sides to the power of $\frac{4}{3}$.

$$x = \left(y^{\frac{1}{2}} - 2\right)^{\frac{4}{3}}$$

3. **Question:** $5x^8 - y = 0$

Solution: Add y to both sides, divide by 5, then take the 8^{th} root.

$$5x^8 = y$$

$$x^8 = \frac{y}{5}$$

$$x = \left(\frac{y}{5}\right)^{\frac{1}{8}} = \sqrt[8]{\frac{y}{5}}$$

4. **Question:** $x^{\frac{3}{4}}y = 8my$

Solution: Divide both sides by y (assuming $y \neq 0$), then raise to the power of $\frac{4}{3}$.

$$x^{\frac{3}{4}} = 8m$$

$$x = (8m)^{\frac{4}{3}}$$

5. **Question:** $2x^{2.5} = 64$

Solution: Divide by 2, then raise to the power of $\frac{1}{2.5}$ (or $\frac{2}{5}$).

$$x^{2.5} = 32$$

$$x^{5/2} = 32$$

$$x = (32)^{2/5} = (\sqrt[5]{32})^2 = 2^2 = 4$$

6. **Question:** $x^2y^3x^5y^2 = a$

Solution: Combine like terms using $x^ax^b = x^{a+b}$.

$$x^{2+5}y^{3+2} = a$$

$$x^7y^5 = a$$

$$x^7 = \frac{a}{y^5}$$

$$x = \left(\frac{a}{y^5}\right)^{\frac{1}{7}} = \frac{a^{\frac{1}{7}}}{y^{\frac{5}{7}}}$$

7. **Question:** $0.5x^{-0.5}y^{-0.5} \times 0.5x^{-0.5}y^{0.5} = 4$

Solution: Multiply the coefficients and combine like terms.

$$(0.5 \times 0.5)x^{-0.5-0.5}y^{-0.5+0.5} = 4$$

$$0.25x^{-1}y^0 = 4$$

$$0.25x^{-1} = 4$$

$$\frac{0.25}{x} = 4$$

$$0.25 = 4x$$

$$x = \frac{0.25}{4} = \frac{1}{16}$$

8. **Question:** $0.8y^{-0.2}x^{0.2} \times 0.2y^{0.8}x^{-0.8} = 10$

Solution: Multiply the coefficients and combine like terms.

$$(0.8 \times 0.2)y^{-0.2+0.8}x^{0.2-0.8} = 10$$

$$0.16y^{0.6}x^{-0.6} = 10$$

$$\begin{aligned}
 x^{-0.6} &= \frac{10}{0.16y^{0.6}} \\
 x^{0.6} &= \frac{0.16y^{0.6}}{10} \\
 x &= \left(\frac{0.16y^{0.6}}{10} \right)^{\frac{1}{0.6}} = \left(\frac{0.16y^{0.6}}{10} \right)^{\frac{5}{3}} \\
 x &= \frac{(0.16)^{5/3}y}{(10)^{5/3}}
 \end{aligned}$$

9. **Question:** $\frac{\alpha y^{\alpha-1} x^{\beta}}{\beta y^{\alpha} x^{\beta-1}} = k$

Solution: Simplify terms using index laws $\frac{A^m}{A^n} = A^{m-n}$.

$$\frac{\alpha}{\beta} y^{\alpha-1-\alpha} x^{\beta-(\beta-1)} = k$$

$$\frac{\alpha}{\beta} y^{-1} x^1 = k$$

$$\frac{\alpha x}{\beta y} = k$$

$$\alpha x = k\beta y$$

$$x = \frac{k\beta y}{\alpha}$$

Multivariate Optimisation Rearranging Practice - Solutions

Zy is a pet dog, who gains utility from eating and from playing catch. Her utility function is defined as:

$$U(E, C) = E^{0.6} C^{0.4}$$

Where E is the number of hours eating and C is the number of hours playing catch.

During the day, Zy has a maximum of 8 hours for eating and catching.

For maximum utility, how many hours should Zy spend eating and catching, to the nearest minute?

Part of a solution:

Constraint function:

$$E + C = 8$$

Lagrangian:

$$L = E^{0.6}C^{0.4} - \lambda(8 - \dots)$$

1. **Question:** Replace the dots with the rearranged constraint function.

Solution: Rearrange the constraint function to be equal to zero: $8 - E - C = 0$.

$$L = E^{0.6}C^{0.4} - \lambda(8 - E - C)$$

First Order Conditions:

$$\frac{dL}{dE} = 0.6E^{-0.4}C^{0.4} - \lambda = 0$$

$$\frac{dL}{dC} = 0.4E^{0.6}C^{-0.6} - \lambda = 0$$

2. **Question:** For both first order conditions, rearrange to make λ the subject of an equation in E and C .

Solution:

$$0.6E^{-0.4}C^{0.4} = \lambda$$

$$0.4E^{0.6}C^{-0.6} = \lambda$$

Dividing the Equations:

You then see that they have divided these two equations by one another:

$$\frac{0.6E^{-0.4}C^{0.4}}{0.4E^{0.6}C^{-0.6}} = \frac{\lambda}{\lambda}$$

3. **Question:** Make C the subject of this equation. You will need to use rules of indices.

Solution: Simplify the numerical coefficients and combine the E and C terms using

index laws.

$$\begin{aligned}\frac{0.6}{0.4}E^{-0.4-0.6}C^{0.4-(-0.6)} &= 1 \\ 1.5E^{-1}C^1 &= 1 \\ \frac{1.5C}{E} &= 1 \\ 1.5C &= E \\ C &= \frac{E}{1.5} = \frac{2}{3}E\end{aligned}$$

Tip: Remember that $\frac{x^a}{x^b} = x^{a-b}$ and $\frac{1}{x^n} = x^{-n}$.

4. **Question:** Substitute your equation for C into the constraint function $E + C = 8$.

Solution: Substitute $C = \frac{2}{3}E$ into the constraint function.

$$E + \frac{2}{3}E = 8$$

$$\frac{3}{3}E + \frac{2}{3}E = 8$$

$$\frac{5}{3}E = 8$$

5. **Question:** Solve this equation for E .

Solution:

$$E = 8 \times \frac{3}{5} = \frac{24}{5} = 4.8$$

6. **Question:** Hence find the values of E and C that maximise utility for Zy and convert your answer to hours and minutes (e.g., 3.5 hours is 3 hours and 30 minutes).

Solution: Using $E = 4.8$ hours, find C :

$$C = \frac{2}{3}E = \frac{2}{3}(4.8) = 2 \times 1.6 = 3.2$$

Convert to hours and minutes: For E : 4.8 hours = 4 hours and $0.8 \times 60 = 48$ minutes. So, 4 hours and 48 minutes. For C : 3.2 hours = 3 hours and $0.2 \times 60 = 12$ minutes. So, 3 hours and 12 minutes.

Final Answer: Zy should spend $E = 4$ hours and 48 minutes eating, and $C = 3$ hours and 12 minutes playing catch.

Challenge Problem: Multivariate Optimisation Question - Solutions

This last question uses more skills than this workshop but is made available for further practice of ECO-5007A work.

A utility function is given:

$$U = x^{\frac{1}{2}}y^{\frac{1}{2}}$$

With constraint:

$$5x + 4y = 20$$

Find values for x and y that maximise the utility function.

Solution:

Lagrangian:

$$L(x, y, \lambda) = x^{\frac{1}{2}}y^{\frac{1}{2}} - \lambda(20 - 5x - 4y)$$

First Order Conditions:

$$\frac{\partial L}{\partial x} = \frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}} - 5\lambda = 0 \Rightarrow \frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}} = 5\lambda \quad (1)$$

$$\frac{\partial L}{\partial y} = \frac{1}{2}x^{\frac{1}{2}}y^{-\frac{1}{2}} - 4\lambda = 0 \Rightarrow \frac{1}{2}x^{\frac{1}{2}}y^{-\frac{1}{2}} = 4\lambda \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = -(20 - 5x - 4y) = 0 \Rightarrow 5x + 4y = 20 \quad (3)$$

Dividing (1) by (2):

$$\frac{\frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}}}{\frac{1}{2}x^{\frac{1}{2}}y^{-\frac{1}{2}}} = \frac{5\lambda}{4\lambda}$$

$$x^{-\frac{1}{2}-\frac{1}{2}}y^{\frac{1}{2}-(-\frac{1}{2})} = \frac{5}{4}$$

$$x^{-1}y^1 = \frac{5}{4}$$

$$\frac{y}{x} = \frac{5}{4}$$

$$y = \frac{5}{4}x$$

Substitute y into the budget constraint (3):

$$5x + 4\left(\frac{5}{4}x\right) = 20$$

$$5x + 5x = 20$$

$$10x = 20$$

$$x = 2$$

Find y :

$$y = \frac{5}{4}(2) = \frac{10}{4} = 2.5$$

Final Answer: Optimal quantities: $x = 2, y = 2.5$